

ORBIT

User Manual

Ordered workflow, complete documented command reference, operational warnings, troubleshooting, and a start-to-finish worked example

Oxidation-state Retrieval Band-gap Insulating path Tracer

Manual basis: the supplied ORBIT README command interface. Use the installed command's --help output as the final authority after a software update.

Prepared September 2026

Contents

1. Purpose and workflow model	4
1.1 Ordered stages	4
1.2 Global command behavior	4
2. Requirements, installation, and resources	5
2.1 Requirements	5
2.2 Install or update	5
2.3 Pseudopotential layout	5
3. Step 1 - Initialize a project	6
3.1 Syntax and modes	6
3.2 Examples	6
3.3 Expected outputs	6
4. Step 2 - Configure orbit.toml	7
4.1 Important settings	8
4.2 Automatic band count	8
4.3 Optional Hubbard U	8
5. Steps 3-4 - Inspect the structure and select targets	9
5.1 Inspect	9
5.2 List targets	9
5.3 Set targets	9
6. Step 5 - Build displacement samples	10
7. Step 6 - Prepare and run sampled SCFs	11
7.1 Outputs to inspect	11
8. Steps 7-8 - Extract and inspect sampled gaps	12
8.1 Extract	12
8.2 Plot sampled gaps	12
9. Step 9 - Construct the winding path	13
10. Steps 10-11 - Calculate and verify path images	14
10.1 Prepare or submit path SCFs	14
10.2 Extract verified path gaps	14

10.3 Plot the combined viewer	14
11. Step 12 - Helper-assisted path refinement	15
11.1 Configure helper search	15
11.2 One helper iteration	15
11.3 Helper command reference	16
11.4 Selection and interpolation behavior	16
12. Steps 13-14 - Response and polarization analysis	17
12.1 Combined response preparation	17
12.2 Independent response commands	17
12.3 Analyze polarization and transported charge	18
13. Files, provenance, restart, and overwrite rules	19
13.1 Compact project layout	19
13.2 Persistence rules	19
14. Warnings, edge cases, common errors, and fixes	20
14.1 High-risk actions	21
15. Worked example - one material from CIF to transported charge	21
15.1 Create and configure the project	21
15.2 Inspect and select the Ti site	21
15.3 Sample and calculate the displacement grid	22
15.4 Construct and calculate the target-only path	22
15.5 Run one or more helper iterations	22
15.6 Prepare final response calculations	23
15.7 Analyze and archive	23
16. Command index	24
16.1 Interface verification after updates	24

1. Purpose and workflow model

ORBIT constructs a periodic atomic-displacement path for a selected crystallographic site, verifies the electronic gap at every path image, optionally refines a non-insulating path with nearby helper atoms, and prepares polarization-response calculations for transported-charge analysis.

1.1 Ordered stages

Stage	Command family	Result
1. Initialize	orbit init	Create the project layout and starter configuration.
2. Configure	Edit orbit.toml	Define structure, sampling, QE, pathfinding, helper, pseudopotential, and Slurm settings.
3. Inspect	orbit inspect	Bind manifests to the CIF and assign stable site IDs.
4. Select targets	orbit targets	Choose the symmetry-inequivalent sites to displace.
5. Sample	orbit sample	Build the symmetry-reduced periodic displacement grid.
6. Sample SCFs	orbit scf	Prepare and optionally submit QE calculations at sampled nodes.
7. Extract	orbit extract	Create an auditable sampled-gap field.
8. Inspect samples	orbit plot	Visualize calculated representatives and symmetry images.
9. Find path	orbit path	Find the winding route with the largest bottleneck gap.
10. Path SCFs	orbit path-scf	Calculate every interpolated path image.
11. Verify path	orbit path-extract / plot-path	Measure and inspect the true minimum path gap.
12. Refine	orbit helper	Add smooth helper-atom motion when required.
13. Respond	orbit path-ph / path-polarization	Calculate response quantities along the accepted path.
14. Analyze	orbit path-response-plot	Select polarization branches and determine transported charge.

Scientific distinction

The sampled graph only proposes a path. The final insulating classification must use the explicitly calculated gap of every interpolated path image; ORBIT does not invent or interpolate uncalculated gaps.

1.2 Global command behavior

- `--root PROJECT_DIRECTORY` is accepted by workflow commands and selects the directory containing `orbit.toml`; the default is the current directory.
- Repeatable `--target SITE_ID` options restrict work to configured targets. If omitted, the command processes all configured targets.
- `--check-only` validates a stage without writing stage outputs.
- Only `--submit` permits ORBIT to invoke `sbatch`.
- `--overwrite` authorizes a conflicting stage refresh; it is not a request to erase the whole project.
- Expected configuration or user errors print `ERROR: ...` and return exit status 2.

Before every large calculation

Run the stage's check-only mode where available, inspect the reported calculation count and resources, and submit only after the validation is clean.

2. Requirements, installation, and resources

2.1 Requirements

- Python 3.11 or newer.
- Quantum ESPRESSO `pw.x`; `ph.x` is needed for Born-effective-charge response calculations.
- A Slurm cluster configured to launch with `mpirun` or `srun`.
- Exactly one selected UPF pseudopotential per element and cutoff metadata in `cutoffs.json`.
- Python dependencies installed with the package, including ASE, NumPy, Plotly, and spglib.

2.2 Install or update

```
cd /path/to/orbit
python -m pip install -e .
orbit --version
orbit --help
```

An editable installation makes later changes under `src/orbit/` visible without reinstalling. In a new shell, activate the Python or Conda environment that contains ORBIT before using the command.

2.3 Pseudopotential layout

```
pseudo/
|-- cutoffs.json
`-- library/
    |-- H....upf
    `-- O....upf

{
  "H": {"cutoff_wfc": 30, "cutoff_rho": 240},
  "O": {"cutoff_wfc": 45, "cutoff_rho": 540}
}
```

ORBIT automatically selects an `Element*.upf` match only when it is unique. If multiple files match an element, name the desired file explicitly under `[pseudopotentials.files]`.

```
[pseudopotentials]
root = "../pseudo"

[pseudopotentials.files]
H = "H.us.pbe.upf"
O = "O.paw.pbe.upf"
```

Pseudopotential consistency

Use cutoff values and exchange-correlation choices appropriate to the exact UPF files. Pseudopotentials are not distributed with ORBIT. Changing a UPF or cutoff after preparation creates a preparation conflict that must be reviewed and explicitly refreshed.

3. Step 1 - Initialize a project

Initialization creates the standard directories and writes a starter configuration when one is absent. It does not inspect the CIF, sample structures, prepare QE inputs, or submit jobs.

3.1 Syntax and modes

```
orbit init --structure STRUCTURE_PATH [--root PROJECT_DIRECTORY] [--force-config]
```

Argument	Requirement	Meaning
--structure STRUCTURE_PATH	Required	Structure path stored under [project].structure. It may be absolute or relative to the project root.
--root PROJECT_DIRECTORY	Optional	Directory to create or reuse. Default: current directory.
--force-config	Optional, destructive to config	Replace an existing orbit.toml with the starter configuration. Other project files are retained.

3.2 Examples

```
orbit init --root /path/to/project --structure crystal.cif
orbit init --root . --structure crystal.cif
orbit init --root . --structure crystal.cif --force-config
```

Existing configuration

Initialization preserves an existing orbit.toml. Use --force-config only when replacing that file is intentional; save a copy first if it contains tuned settings.

3.3 Expected outputs

The project receives orbit.toml when needed and the standard manifests/, samples/, calculations/, results/, paths/, plots/, and logs/ directories.

4. Step 2 - Configure orbit.toml

Edit the configuration before inspection and calculation. The following compact template contains the core settings documented by ORBIT.

```
schema_version = 1

[project]
structure = "crystal.cif"

[sampling]
spacing_angstrom = 0.30
fold_by_symmetry = true
symprec_angstrom = 0.001
overlap_tolerance_angstrom = 1.0e-5

[targets]
site_ids = []

[pathfinding]
theta_min_eV = 0.10
n_images = 50
winding_padding = 1

[qe]
kpoints = [6, 6, 6]
offsets = [0, 0, 0]
conv_thr = 1.0e-10
mixing_beta = 0.30
occupations = "fixed"
exact_overlap_tolerance_angstrom = 1.0e-5

[pseudopotentials]
root = "../pseudo"

[slurm]
partition = "regular"
nodes = 1
tasks_per_node = 48
time = "02:00:00"
max_concurrent = 6
modules = ["espresso/module"]
launcher = "mpirun"
pw_command = "pw.x"
```

4.1 Important settings

Setting	Effect and caution
sampling.spacing_angstrom	Approximate Cartesian target-grid spacing. Smaller spacing increases sample count and cost, often sharply.
sampling.fold_by_symmetry	Retain one representative per orbit of the target-site stabilizer.
sampling.symprec_angstrom	spglib symmetry tolerance. An unsuitable value can merge or split site orbits.
sampling.overlap_tolerance_angstrom	Sampling-stage direct nuclear-coincidence tolerance.
pathfinding.theta_min_eV	Required path bottleneck threshold. The documented success test is strictly greater than this value.
pathfinding.n_images	Number of interpolated path structures; must be at least 2.
pathfinding.winding_padding	Extra periodic cells made available to the winding search; nonnegative integer.
qe.kpoints / offsets	Monkhorst-Pack grid and offsets.
qe.nbnd	Optional explicit band count. If absent, ORBIT derives it from UPF valence counts.
qe.exact_overlap_tolerance_angstrom	Final geometry check before a QE input is written.
slurm.max_concurrent	Maximum simultaneously active tasks in each array.
slurm.launcher	Must be mpirun or srun.

4.2 Automatic band count

With automatic nbnd, ORBIT uses $N_{\text{occ}} = N_{\text{e}}/2$, where N_{e} is the neutral-cell UPF valence-electron count and N_{occ} is the occupied-band count. It adds at least eight empty bands, or 25% of N_{occ} for larger systems, and rounds upward to a multiple of eight.

Automatic nbnd limitation

Automatic setup requires an even valence-electron count and does not configure spin polarization. Odd-electron, magnetic, charged, smearing-based, or otherwise nonstandard calculations require deliberate extension or manual configuration and are outside the documented default scope.

4.3 Optional Hubbard U

```
[qe.hubbard]
projector = "atomic"
```

```
[[qe.hubbard.u]]
element = "0"
manifold = "2p"
value_eV = 6.0
```

Treat the element, manifold, projector, and U value as physical model choices. The example is syntax, not a recommendation for a particular material.

5. Steps 3-4 - Inspect the structure and select targets

5.1 Inspect

```
orbit inspect [--root PROJECT_DIRECTORY]
```

The command reads the configured structure, detects symmetry, assigns stable site IDs, and writes manifests/structure.json and manifests/sites.csv. The terminal reports the formula, volume, space group, site IDs, site orbits, and fractional coordinates.

CIF changes invalidate provenance

Always rerun inspection after changing the CIF. Downstream stages compare hashes and reject stale manifests. Review stable site IDs again before reselecting targets.

5.2 List targets

```
orbit targets list [--root PROJECT_DIRECTORY]
```

This read-only command displays every inspected site and marks current targets. It verifies that configured target IDs still exist.

5.3 Set targets

```
orbit targets set [--root PROJECT_DIRECTORY] \
  (--all-inequivalent | --elements ELEMENT [ELEMENT ...] | --sites SITE_ID [SITE_ID ...])
```

Mode	Meaning
--all-inequivalent	Select one representative from every crystallographic site orbit.
--elements ELEMENT ...	Select all inequivalent sites belonging to the listed case-sensitive chemical elements.
--sites SITE_ID ...	Select explicit stable IDs in the supplied order.

```
orbit targets set --all-inequivalent
orbit targets set --elements 0 Ti
orbit targets set --sites 01 H1 H2
orbit targets list
```

Selection modes

Exactly one selection mode is required. Modes cannot be combined, and running targets set again replaces rather than appends to the previous target list.

6. Step 5 - Build displacement samples

```
orbit sample [--root PROJECT_DIRECTORY] [--target SITE_ID ...] \  
  [--check-only | --overwrite]
```

Argument	Meaning
--root PROJECT_DIRECTORY	Project directory; default is the current directory.
--target SITE_ID	Repeatable restriction to configured targets; omit for all targets.
--check-only	Compute and validate grid sizes, representatives, valid geometries, and overlap walls without writing the dataset.
--overwrite	Atomically replace a conflicting target sampling dataset.

```
orbit sample --check-only  
orbit sample  
orbit sample --target 01 --check-only  
orbit sample --target 01  
orbit sample --target 01 --target H1  
orbit sample --target 01 --overwrite
```

Outputs are written below `samples/TARGET/`: `sampling.json`, `samples.csv`, and valid CIFs in `structures/`. Direct-overlap rows remain in the CSV with status `DIRECT_OVERLAP` but receive no CIF and become impassable pathfinding walls.

Cost scaling

Inspect the check-only counts before writing or submitting. Finer grids increase both storage and SCF count. Symmetry folding reduces redundant calculations but does not relax geometry checks.

Invalid combination

--check-only and --overwrite cannot be combined. An identical complete sampling dataset is skipped; a conflicting or incomplete one requires explicit overwrite.

7. Step 6 - Prepare and run sampled SCFs

```
orbit scf [--root PROJECT_DIRECTORY] [--target SITE_ID ...] \
  [--check-only] [--overwrite] [--submit] [--max-concurrent COUNT]
```

Argument	Meaning
--target SITE_ID	Repeatable target restriction; omit for all configured targets.
--check-only	Validate UPFs, cutoffs, electron and band counts, geometries, QE rendering, and Slurm settings without writing or submitting.
--overwrite	Refresh conflicting prepared inputs. Changed outputs and tmp directories are preserved with stale timestamp names.
--submit	Prepare, then execute each target's submit script. Only unfinished valid calculations are submitted.
--max-concurrent COUNT	Override the configured positive Slurm array concurrency cap.

```
orbit scf --target 01 --check-only
orbit scf --target 01
orbit scf --target 01 --max-concurrent 8 --submit
```

```
# Monitor outside ORBIT
squeue -u "$USER"
```

```
# Safe resume of unfinished calculations
orbit scf --target 01 --submit
```

Prepared files live under `calculations/samples/TARGET/`. The restart-aware list omits invalid rows and outputs that already contain `JOB DONE`.

No implicit submission

Running `orbit scf` without `--submit` writes inputs and scripts but does not call `sbatch`. Review generated inputs and the valid-calculation list before first submission.

Invalid combinations

Do not combine `--check-only` with `--submit` or `--overwrite`. Use `--overwrite` only after identifying why the prepared inputs conflict with current settings.

7.1 Outputs to inspect

- `preparation.json`: preparation provenance and settings.
- `calculations.csv`: complete calculation inventory and statuses.
- `valid_calculations.txt`: entries eligible for the array.
- `run_array.sh` and `submit.sh`: generated execution and submission scripts.
- `sample_*/espresso_scf.pwi`: individual QE inputs.

8. Steps 7-8 - Extract and inspect sampled gaps

8.1 Extract

```
orbit extract [--root PROJECT_DIRECTORY] [--target SITE_ID ...]
```

```
orbit extract
```

```
orbit extract --target 01
```

```
orbit extract --target 01 --target H1
```

Extraction parses every authoritative sample into `results/gaps/TARGET/gaps.csv` and `extraction.json`. Each sample remains visible even if its output is missing, incomplete, unreadable, invalid, or gapless. A row is complete only when QE provides a usable HOMO/LUMO pair and JOB DONE.

Partial extraction

Do not treat a partial sampled-gap table as a complete pathfinding field. Resume missing SCFs and extract again. Missing and failed nodes become walls; enough of them can make a real path appear infeasible.

8.2 Plot sampled gaps

```
orbit plot [--root PROJECT_DIRECTORY] [--target SITE_ID ...] \
  [--output HTML_PATH] [--representatives-only]
```

Argument	Meaning
--target SITE_ID	Repeatable; multiple targets appear as tabs with a shared gap-color range.
--output HTML_PATH	Select the standalone HTML output path; relative paths resolve from the project root.
--representatives-only	Hide symmetry-equivalent images and show only calculated representatives.

```
orbit plot --target 01
```

```
orbit plot --target 01 --representatives-only
```

```
orbit plot --target 01 --target H1 --target H2
```

```
orbit plot --target 01 --output plots/01_custom.html
```

Plotting writes a standalone HTML visualization and same-basename JSON manifest. It never submits or reruns QE.

9. Step 9 - Construct the winding path

ORBIT evaluates the seven positive lattice windings a , b , c , ab , bc , ac , and abc . It searches the periodic sampled graph for the route that maximizes the minimum gap along the route.

```
orbit path [--root PROJECT_DIRECTORY] [--target SITE_ID ...] \
  [--theta-min EV] [--n-images COUNT] [--winding-padding COUNT] [--no-cifs]
```

Argument	Meaning
--target SITE_ID	Repeatable target restriction; each target receives a separate immutable run.
--theta-min EV	Override the configured threshold. The documented success test requires the selected bottleneck to strictly exceed it.
--n-images COUNT	Override the number of path images; integer at least 2.
--winding-padding COUNT	Override periodic search-box padding; nonnegative integer.
--no-cifs	Record diagnostic pathfinding without writing path CIFs. This run cannot proceed to path-scf.

```
orbit path --target 01
orbit path --target 01 --theta-min 0.10 --n-images 50
orbit path --target 01 --winding-padding 2
orbit path --target 01 --no-cifs
```

Every attempt receives a unique directory under `paths/TARGET/runs/RUN_ID/`, including failed attempts. Successful runs normally contain `run.json`, `candidates.csv`, `pathfinding.log`, `path.csv`, `path.json`, and `cifs/`.

Infeasible is diagnostic, not destructive

An INFEASIBLE result does not erase earlier runs. Inspect `candidates.csv` and `pathfinding.log`. Confirm sampled SCFs and extraction are complete before changing resolution, geometry rules, padding, or the gap threshold.

Do not confuse sampled and verified gaps

The graph bottleneck is based on sampled nodes. Interpolation between nodes may introduce a lower gap, so path-image SCFs are mandatory before declaring the trajectory insulating.

10. Steps 10-11 - Calculate and verify path images

10.1 Prepare or submit path SCFs

```
orbit path-scf [--root PROJECT_DIRECTORY] [--target SITE_ID ...] \
  [--run-id RUN_ID] [--check-only] [--overwrite] [--submit] \
  [--max-concurrent COUNT]
```

Argument	Meaning
--run-id RUN_ID	Pin a historical run. Allowed only with exactly one target. Otherwise the latest successful path is selected.
--check-only	Validate the run, resources, all CIFs and geometries, QE input rendering, and Slurm settings.
--overwrite	Refresh conflicting preparation while preserving changed outputs and tmp directories as stale copies.
--submit	Submit only unfinished valid images.
--max-concurrent COUNT	Override the positive array concurrency cap.

```
orbit path-scf --target 01 --run-id RUN_ID --check-only
orbit path-scf --target 01 --run-id RUN_ID
orbit path-scf --target 01 --run-id RUN_ID --submit
```

```
# Safe resume
orbit path-scf --target 01 --run-id RUN_ID --submit
```

Run selection

Pin --run-id for reproducibility. Without it, ORBIT selects the latest successful run. A diagnostic --no-cifs run cannot be prepared.

10.2 Extract verified path gaps

```
orbit path-extract [--root PROJECT_DIRECTORY] [--target SITE_ID ...] \
  [--run-id RUN_ID]

orbit path-extract --target 01
orbit path-extract --target 01 --run-id RUN_ID
orbit path-extract --target 01 --target H1
```

Outputs are written to results/path_gaps/TARGET/RUN_ID/. Missing or incomplete images remain explicit rows. Finish or resubmit them and re-extract before interpreting the minimum gap.

10.3 Plot the combined viewer

```
orbit plot-path [--root PROJECT_DIRECTORY] [--target SITE_ID ...] \
  [--run-id RUN_ID] [--output HTML_PATH] [--representatives-only]

orbit plot-path --target 01 --run-id RUN_ID
orbit plot-path --target 01 --representatives-only
orbit plot-path --target 01 --target H1 --target H2
orbit plot-path --target 01 --output plots/01_final.html
```

The viewer combines the sampled heatmap, trajectory, calculated path-gap curve, structure, and controls. It supports heatmap and path modes, manual step selection, play/pause, sampled-point visibility, and light/dark themes.

11. Step 12 - Helper-assisted path refinement

Use helper refinement when the explicitly calculated target-only path does not meet the configured insulating threshold. ORBIT identifies the current minimum-gap image, scans nearby atoms independently, selects a geometry-safe electronic improvement, and adds a smooth displacement field while retaining the target atom's original path.

11.1 Configure helper search

```
[helper]
target_gap_eV = 0.15
neighbor_cutoff_angstrom = 3.50
scan_axes = "xyz"
scan_radius_angstrom = 0.30
scan_steps_per_axis = 5
minimum_helper_distance_angstrom = 1.20
taper_half_width_images = 8.0
improvement_tolerance_eV = 1.0e-6
```

Setting	Meaning
target_gap_eV	Minimum calculated gap required at every refined path image.
neighbor_cutoff_angstrom	Periodic minimum-image radius used to find helper candidates.
scan_axes	Cartesian displacement axes included in each independent helper scan.
scan_radius_angstrom	Maximum scan displacement scale.
scan_steps_per_axis	Grid resolution along each enabled axis.
minimum_helper_distance_angstrom	Additional geometry-safety floor for helper corrections; 0 disables this floor but not exact-overlap validation.
taper_half_width_images	Width over which helper motion smoothly returns toward zero.
improvement_tolerance_eV	Electronic-improvement comparison tolerance.

11.2 One helper iteration

```
orbit helper status --target Ti1 --run-id RUN_ID

orbit helper scan --target Ti1 --run-id RUN_ID --check-only
orbit helper scan --target Ti1 --run-id RUN_ID --submit

# Wait for all helper-scan SCFs.
orbit helper analyze --target Ti1 --run-id RUN_ID
orbit helper path-scf --target Ti1 --run-id RUN_ID --submit

# Wait for all refined-path SCFs.
orbit helper extract --target Ti1 --run-id RUN_ID
orbit helper plot --target Ti1 --run-id RUN_ID
orbit helper status --target Ti1 --run-id RUN_ID
```

11.3 Helper command reference

Command	Documented modes and effect
orbit helper status	Inspect the campaign, resolved path run, iteration state, bottleneck, anchors, and convergence state. Accepts target and optional run ID.
orbit helper scan	Prepare candidate helper scans at the current bottleneck. --check-only validates/counts; --submit submits. --force bypasses the normal review gate for expert use.
orbit helper analyze	Read completed candidate scans, select the best safe improvement, record or update the anchor, and construct the next refined path.
orbit helper path-scf	Prepare and optionally submit SCFs for every image of the refined path.
orbit helper extract	Extract calculated gaps for the current helper iteration and classify its state.
orbit helper plot	Plot helper iterations and, for a nonconverged reviewed iteration, mark it READY_FOR_NEXT.

Normal helper commands may omit --run-id. ORBIT first reuses the newest helper campaign for the target; if none exists, it uses the newest constructed path run. The resolved run is printed. Specify the run ID when exact provenance matters.

11.4 Selection and interpolation behavior

- The scan finds neighboring atoms by periodic minimum-image distance and scans candidates independently on a Cartesian grid.
- Analysis selects one helper atom at the point with the largest calculated gap; smaller displacement is the tie-breaker.
- The winner is stored as an anchor relative to the immutable target-only reference path.
- Existing anchors persist. Returning to the same bottleneck image updates that anchor, allowing additional helper atoms to accumulate there.
- Cubic smoothstep interpolation joins anchors; a raised-cosine taper returns the correction to zero outside the outermost anchors.

Manual gate

After extraction an iteration is CONVERGED only when every calculated gap reaches target_gap_eV. Otherwise it becomes NEEDS_REVIEW. Plot/review changes a nonconverged iteration to READY_FOR_NEXT. The next scan is blocked until that checkpoint unless --force is used intentionally.

Geometry floor nuance

A pre-existing short contact is not required to become longer merely to start an iteration. The helper correction is forbidden from making that contact still shorter. Inspect all extreme displacements even when the numerical floor passes.

12. Steps 13-14 - Response and polarization analysis

Begin response calculations only after every SCF image of the accepted target or helper path is complete. Response preparation reuses each image's completed ground-state tmp/PREFIX.save directory.

12.1 Combined response preparation

```
orbit path-response --target T1l --run-id RUN_ID \
  --source path --check-only
orbit path-response --target T1l --run-id RUN_ID \
  --source path --submit

orbit path-response --target T1l --run-id RUN_ID \
  --source helper --iteration ITERATION --check-only
orbit path-response --target T1l --run-id RUN_ID \
  --source helper --iteration ITERATION --submit
```

Argument/mode	Meaning
--target SITE_ID	Select the configured target.
--run-id RUN_ID	Pin the underlying target-only path campaign.
--source path helper	Choose the accepted target-only path or a helper-refined path.
--iteration COUNT	Pin a helper iteration when source is helper; otherwise the latest ready iteration may be selected.
--check-only	Validate response prerequisites and planned jobs without submission.
--submit	Prepare and submit the selected response families.
--kind ph polarization	Restrict the backward-compatible combined command to one response family.

12.2 Independent response commands

Independent commands isolate files, manifests, job lists, scripts, and Slurm logs below response/ph or response/polarization.

```
orbit path-ph --target T1l --run-id RUN_ID \
  --source helper --iteration ITERATION --check-only
orbit path-ph --target T1l --run-id RUN_ID \
  --source helper --iteration ITERATION --overwrite --submit

orbit path-polarization --target T1l --run-id RUN_ID \
  --source helper --iteration ITERATION --check-only
orbit path-polarization --target T1l --run-id RUN_ID \
  --source helper --iteration ITERATION --overwrite --submit
```

The polarization command has no PH dependency; it reuses the ground-state SCF saved state directly. Both submitters skip completed outputs. Polarization submission resumes missing directions without rerunning completed ones.

Saved-state prerequisite

Response submission is refused if any required SCF output or PREFIX.save directory is missing. Complete the path or helper SCFs first; do not copy saved states between mismatched images.

12.3 Analyze polarization and transported charge

```
orbit path-response-plot --target Ti1 --run-id RUN_ID \
  --source helper --iteration ITERATION
```

The analysis parses Born tensors, uses all atoms in the incremental response estimate, selects the nearest Berry branch, derives the transport vector from `path.json`, and writes standalone Plotly HTML files plus CSV audit tables under `response/analysis`.

The incremental response relation is $dP = (e/\Omega) \sum_{\kappa} Z_{\kappa}^* du_{\kappa}$, where P is polarization, e is the positive elementary charge, Ω is cell volume, Z_{κ}^* is atom κ 's Born effective-charge tensor, and du_{κ} is its displacement vector.

The transported charge is $N = (\Omega/e) [\Delta P \cdot R] / |R|^2$, where N is the transported charge in units of e , ΔP is the continuous polarization change, and R is the saved lattice winding translation. ORBIT reads R from path provenance rather than assuming a Cartesian axis.

Branch continuity requires insulation

A metallic image breaks the continuous insulating Berry-polarization construction. Do not interpret a transported integer from a path whose calculated minimum gap fails the accepted criterion.

13. Files, provenance, restart, and overwrite rules

13.1 Compact project layout

```
project/
|-- crystal.cif
|-- orbit.toml
|-- manifests/           # structure, sites, targets
|-- samples/TARGET/      # grid metadata and sampled CIFs
|-- calculations/
|   |-- samples/TARGET/   # QE inputs, arrays, outputs
|   |-- paths/TARGET/RUN_ID/
|-- results/
|   |-- gaps/TARGET/
|   |-- path_gaps/TARGET/RUN_ID/
|-- paths/TARGET/runs/RUN_ID/
|-- response/            # PH, polarization, analysis
|-- plots/
`-- logs/
```

13.2 Persistence rules

- An identical completed preparation is skipped.
- Submission includes only unfinished valid calculations.
- Conflicting sampling or QE preparation requires explicit overwrite.
- Changed QE inputs preserve old outputs and temporary directories as timestamped stale copies.
- Each pathfinding attempt is immutable and receives a new run ID.
- A pinned run ID prevents accidental use of a newer historical path.
- Extracted CSVs preserve non-complete statuses instead of hiding missing data.
- Structure and source hashes prevent downstream reuse of stale artifacts.

Practical provenance rule

Record the target site ID, path run ID, helper iteration, ORBIT version, QE version, pseudopotential set, orbit.toml, and final analysis output together whenever a result is reported or published.

14. Warnings, edge cases, common errors, and fixes

Symptom or risk	Likely cause	Safe response
No targets selected	targets.site_ids is empty or selection was never written.	Run inspect, list sites, select one valid mode with targets set, then list again.
Structure source changed after inspection	CIF bytes differ from the manifest hash.	Rerun inspect, review stable IDs and symmetry, and reselect targets before rebuilding downstream stages.
Multiple pseudopotentials found	More than one Element*.upf file matches.	Specify the exact file under [pseudopotentials.files]; do not let filename order choose the physics.
Preparation conflict	Structure, QE settings, UPF, cutoff, sampling, or source provenance changed.	Inspect the difference. If replacement is intended, rerun the relevant stage with --overwrite; retained stale outputs are not current results.
Jobs absent from the Slurm array	Rows are invalid, rejected, or already contain JOB DONE.	Inspect calculations.csv, valid_calculations.txt, and submission logs. Missing array entries can be expected.
Array finishes but extraction is incomplete	QE terminated early, did not print JOB DONE, or produced no usable HOMO/LUMO pair.	Inspect individual outputs and Slurm logs, fix the cause, safely resubmit unfinished work, then extract again.
Pathfinding is INFEASIBLE	Missing-data walls, overlap walls, coarse topology, insufficient padding, or threshold too high.	First verify complete sampled extraction. Then inspect candidates.csv and pathfinding.log before changing physical or numerical settings.
Path graph succeeds but calculated path dips	Interpolation between sampled nodes introduced a lower-gap configuration.	Use the calculated path result as authoritative. Refine sampling or begin helper refinement; never cite only the graph bottleneck.
path-scf cannot use selected run	The run failed, was created with --no-cifs, or the run ID belongs to another target.	Choose a successful run with CIFs and verify target/run pairing.
run-id rejected	The command received zero or multiple targets with a pinned ID.	Request exactly one target when pinning --run-id.
Automatic nbnd fails	Odd valence-electron count or unsupported spin state.	Set an appropriate explicit band count and validate the physical workflow; default ORBIT does not configure spin polarization.
Helper scan is gated	Previous nonconverged iteration has not been plotted/reviewed.	Run helper plot and status. Use --force only after intentional expert review.
No helper improves the gap	Candidate cutoff/grid is insufficient, safe region contains no improvement, or electronic closure needs collective motion.	Inspect scans and geometry; reconsider cutoff, radius, axes, resolution, and the one-helper-at-a-time assumption.
Helper geometry looks extreme	Electronic winner lies at a scan boundary or sparse grid artifact.	Inspect structures and minimum distances; tighten safety/radius or refine locally rather than accepting the number alone.
Response preparation refuses submission	Path SCF output or PREFIX.save state is missing.	Complete or recover every source image first, then rerun response check-only.
Polarization directions are partially complete	Some directional jobs failed or were never submitted.	Resubmit path-polarization; completed directions are skipped, then verify all enabled directions.
Transported charge is noninteger or unstable	Branch choice, insufficient k-point convergence, metallic images, numerical noise, or incorrect winding provenance.	Verify insulation, convergence, cell volume/units, path winding, and branch continuity before assigning an oxidation state.

Symptom or risk	Likely cause	Safe response
HTML plot fails in one browser	WebGL/browser acceleration or local security behavior.	Try an updated browser with hardware acceleration; preserve the JSON/CSV as the authoritative numerical record.

14.1 High-risk actions

- Do not lower the insulating threshold merely to force a successful status without interpreting the physical meaning.
- Do not use `--force-config`, `--overwrite`, or helper `--force` as routine first responses to an unexplained conflict.
- Do not mix files from different path runs or helper iterations.
- Do not infer completion from an empty queue alone; verify `JOB DONE`, extraction status, and expected row count.
- Do not treat an SCF HOMO-LUMO proxy as a fully converged fundamental band gap without appropriate k-point and electronic-structure validation.

15. Worked example - one material from CIF to transported charge

This example uses a generic cubic SrTiO₃ CIF and target label Ti1. Site IDs depend on the actual CIF and inspection result; replace Ti1 if ORBIT reports another stable ID. Paths and helper behavior are material- and configuration-dependent.

15.1 Create and configure the project

```
mkdir -p /path/to/SrTiO3
cd /path/to/SrTiO3
cp /path/to/input/SrTiO3.cif crystal.cif

orbit init --root . --structure crystal.cif

# Edit orbit.toml: sampling, QE, pseudopotentials, Slurm,
# pathfinding, helper, PH, and polarization settings.
```

15.2 Inspect and select the Ti site

```
orbit inspect
orbit targets list
orbit targets set --sites Ti1
orbit targets list
```

Confirm the structure hash, space group, coordinates, chemical identity, and that Ti1 is the intended crystallographic site.

15.3 Sample and calculate the displacement grid

```
orbit sample --target T1l --check-only
orbit sample --target T1l

orbit scf --target T1l --check-only
orbit scf --target T1l
orbit scf --target T1l --submit

squeue -u "$USER"

# After the array leaves the queue, verify outputs rather than assuming success.
orbit extract --target T1l
orbit plot --target T1l
```

If extraction reports incomplete nodes, inspect outputs and Slurm logs, resubmit safely with `orbit scf --target T1l --submit`, and extract again.

15.4 Construct and calculate the target-only path

```
orbit path --target T1l

# Copy the printed run ID for reproducibility.
RUN_ID=REPLACE_WITH_PRINTED_RUN_ID

orbit path-scf --target T1l --run-id "$RUN_ID" --check-only
orbit path-scf --target T1l --run-id "$RUN_ID"
orbit path-scf --target T1l --run-id "$RUN_ID" --submit

squeue -u "$USER"

orbit path-extract --target T1l --run-id "$RUN_ID"
orbit plot-path --target T1l --run-id "$RUN_ID"
```

Inspect the calculated gap of every image and the minimum-gap structure. If all images meet the accepted threshold, skip directly to response calculations. Otherwise continue with helper refinement.

15.5 Run one or more helper iterations

```
orbit helper status --target T1l --run-id "$RUN_ID"
orbit helper scan --target T1l --run-id "$RUN_ID" --check-only
orbit helper scan --target T1l --run-id "$RUN_ID" --submit

squeue -u "$USER"

orbit helper analyze --target T1l --run-id "$RUN_ID"
orbit helper path-scf --target T1l --run-id "$RUN_ID" --submit

squeue -u "$USER"

orbit helper extract --target T1l --run-id "$RUN_ID"
orbit helper plot --target T1l --run-id "$RUN_ID"
orbit helper status --target T1l --run-id "$RUN_ID"
```

If status is `READY_FOR_NEXT`, repeat scan through plot. If it is `CONVERGED`, record the final helper iteration number. Do not proceed from `NEEDS_REVIEW` without plotting and inspecting the structures.

15.6 Prepare final response calculations

```
ITERATION=REPLACE_WITH_FINAL_HELPER_ITERATION
```

```
orbit path-ph --target Ti1 --run-id "$RUN_ID" \  
  --source helper --iteration "$ITERATION" --check-only  
orbit path-ph --target Ti1 --run-id "$RUN_ID" \  
  --source helper --iteration "$ITERATION" --submit  
  
orbit path-polarization --target Ti1 --run-id "$RUN_ID" \  
  --source helper --iteration "$ITERATION" --check-only  
orbit path-polarization --target Ti1 --run-id "$RUN_ID" \  
  --source helper --iteration "$ITERATION" --submit
```

```
squeue -u "$USER"
```

For an accepted target-only path, replace `--source helper --iteration ...` with `--source path`.

15.7 Analyze and archive

```
orbit path-response-plot --target Ti1 --run-id "$RUN_ID" \  
  --source helper --iteration "$ITERATION"
```

- Verify that every response direction and image completed successfully.
- Inspect the selected continuous Berry branch and the Born-charge-guided incremental estimate.
- Confirm the saved winding translation, cell volume, units, normalization, and final transported charge.
- Archive configuration, versions, run ID, helper iteration, extraction CSVs, plots, and logs with the reported result.

Example status

The example is a command template, not a guarantee that SrTiO₃, Ti1, or any chosen functional will require exactly one helper iteration or yield a particular gap or oxidation state.

16. Command index

Command	Primary function	Writes	Can submit
orbit --help / --version	Display interface or installed version.	No	No
orbit init	Create project and starter configuration.	Yes	No
orbit inspect	Inspect structure and symmetry; assign stable IDs.	Yes	No
orbit targets list	Display sites and target selection.	No	No
orbit targets set	Replace target selection.	Yes	No
orbit sample --check-only	Validate/count samples.	No stage outputs	No
orbit sample	Write sampled structures and topology.	Yes	No
orbit scf --check-only	Validate sampled SCF preparation.	No stage outputs	No
orbit scf	Prepare sampled QE inputs and arrays.	Yes	No
orbit scf --submit	Submit unfinished sampled SCFs.	Yes	Yes
orbit extract	Extract sampled electronic gaps.	Yes	No
orbit plot	Write sampled-gap HTML viewer.	Yes	No
orbit path	Construct immutable winding path run.	Yes	No
orbit path-scf	Prepare path-image QE inputs and arrays.	Yes	No
orbit path-scf --submit	Submit unfinished path images.	Yes	Yes
orbit path-extract	Extract calculated path-image gaps.	Yes	No
orbit plot-path	Write combined heatmap/path viewer.	Yes	No
orbit helper status	Inspect helper campaign state.	No material stage change	No
orbit helper scan	Prepare or submit helper candidate scans.	Yes	With --submit
orbit helper analyze	Select helper and construct refined path.	Yes	No
orbit helper path-scf	Prepare or submit refined-path SCFs.	Yes	With --submit
orbit helper extract	Extract refined-path gaps and classify state.	Yes	No
orbit helper plot	Plot/review iterations and open next gate.	Yes	No
orbit path-response	Combined PH/polarization preparation.	Yes	With --submit
orbit path-ph	Independent PH response preparation.	Yes	With --submit
orbit path-polarization	Independent Berry-polarization preparation.	Yes	With --submit
orbit path-response-plot	Analyze polarization and transported charge.	Yes	No

16.1 Interface verification after updates

```

orbit --version
orbit --help
orbit COMMAND --help
orbit targets --help
orbit targets set --help
orbit helper --help

```

After an update, the installed parser is authoritative; revise this manual together with the CLI and concise README.